Auteur: Oubayy Ahale
Studierichting: Informatica
Oefeningenreeks 7E nr. 1.2

$f(x, y) = x^2 + 2y^2 - 4x - 8y - 5$

$\frac{\partial f}{\partial x} = 2x - 4 = 0 \Leftrightarrow x = 2$

$\frac{\partial f}{\partial y} = 4y - 8 = 0 \Leftrightarrow y = 2$

$H(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$

$H(2, 4) = 8 > 0$

$\frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2} > 0$

$\Rightarrow (2, 2)$ is een minimum

References